Unit 3

Convection
- Mechanism
- thermal and velocity boundary layers
- boundary layer thickness
- relationship between hydrodynamic and thermal boundary layer thickness for flow over flat plates
- the convective heat transfer coefficient, reference temperatures
- thermal boundary layers for the cases of flow over a flat plate and flow through pipe
- dimensionless numbers in heat transfer and their significance

Forced Convection
- General methods for estimation of convection heat transfer coefficient
- Correlation equations for heat transfer in laminar and turbulent flow for external and internal flows for constant heat flux and wall temperature conditions
- flow in a circular tube
- Analogy between momentum and heat transfer: Development of Reynold's and Prandtl analogy

Natural Convection
- Dimensional analysis
- natural convection from vertical and horizontal surfaces under laminar and turbulent conditions for plates, cylinders
- physical significance of Grashoff and Rayleigh numbers

Convection heat transfer

- Convection heat transfer strongly depends on the fluid properties dynamic viscosity, thermal conductivity k, density, and specific heat Cp, as well as the fluid velocity.
- It also depends on the geometry and the roughness of the solid surface, in addition to the type of fluid flow (such as being streamlined or turbulent).
- Thus, convection heat transfer relations are rather complex because of the dependence of convection on so many variables.
- The rate of convection heat transfer is observed to be proportional to the temperature difference and is conveniently expressed by Newton's law of cooling as

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty)$$

- Convection heat transfer is complicated by the fact that it involves fluid motion as well as heat conduction.
- The fluid motion enhances heat transfer, since it brings hotter and cooler chunks of fluid into contact, initiating higher rates of conduction at a greater number of sites in a fluid.
- **Therefore, the rate of heat transfer through a fluid is much higher by convection than it is by conduction.**
- In fact, the higher the fluid velocity, the higher the rate of heat transfer.

Reference Books:
- Cengel A. Yunnus. "*Heat Transfer – A Practical Approach*", McGraw Hill.

h = convection heat transfer coefficient, $W/m^2 \cdot {}^\circ C$

A_s = heat transfer surface area, m^2

T_s = temperature of the surface, $^\circ C$

T_∞ = temperature of the fluid sufficiently far from the surface, $^\circ C$

- Convection heat transfer coefficient can be defined as the rate of heat transfer between a solid surface and a fluid per unit surface area per unit temperature difference.

Convection is classified as:

(i) free or natural convection and
(ii) forced convection.

When the circulating currents arise from the heat transfer process itself, i.e., from the density differences arising in turn due to temperature differences / gradients within the fluid mass, the mode of heat transfer is called free or natural convection.

Examples of natural convection:

1. Heating of a vessel containing liquid by means of a gas flame situated underneath. The liquid at the bottom of the vessel gets heated, expands and rises because its density has become less than that of the remaining liquid. Cold liquid of higher density takes its place and a circulating current is set up.

2. The flow of air across a heated radiator/heat of a room by means of a steam radiator.

When the circulating currents are produced by an external agency such as an agitator in a reaction vessel, pump, fan or blower, the mode of heat transfer is called forced convection. Here fluid motion is independent of density gradients.

Example of forced convection:

Heat flow to a fluid pumped through a heated pipe.

In general, higher rates of heat transfer are obtained in forced convection as compared to natural convection owing to a greater magnitude of circulation in the forced circulation.

Individual and Overall heat transfer coefficients

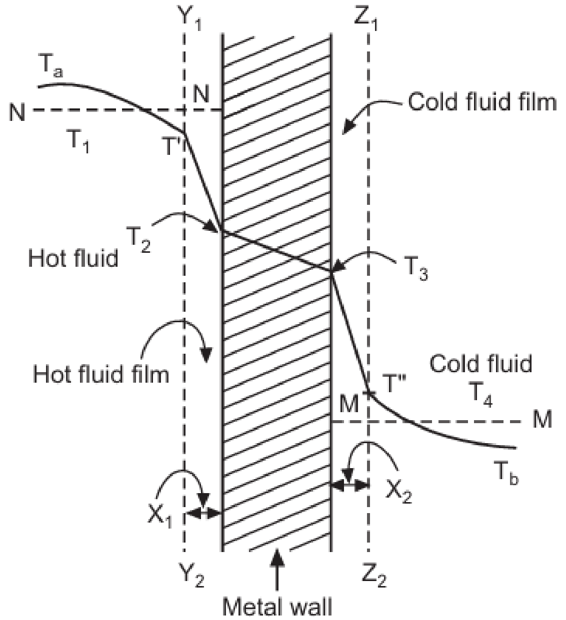


Fig. 3.1 : Temperature gradient in forced convection

The temperature change from T1 to T2 is taking place in the hot fluid film of thickness x1. The rate of heat transfer through this film by conduction is given by

$$Q = \frac{k_1 A_1 (T_1 - T_2)}{x_1}$$

The effective film thickness x1 depends upon the nature of flow and nature of the surface, and is generally not known. Therefore Equation is usually rewritten as

$$Q = h_i A_i (T_1 - T_2)$$

where hi is known as the inside heat transfer coefficient or the surface coefficient or the simply film coefficient.

The overall resistance to heat flow from the hot fluid to the cold fluid is made up of three resistances in series. They are : 1. Resistance offered by the film of hot fluid. 2. Resistance offered by the metal wall and 3. Resistance offered by the film of cold fluid.

The rate of heat transfer through the metal wall is given by :

$$Q = \frac{kA_w (T_2 - T_3)}{x_w}$$

A_w – log mean area of pipe

x_w – thickness of pipe wall

k – thermal conductivity of material of pipe.

The rate of heat transfer through the cold fluid film is given by

$$Q = h_o A_o (T_3 - T_4)$$

where ho is the outside film coefficient or individual heat transfer coefficient.

$$(T_1 - T_2) + (T_2 - T_3) + (T_3 - T_4) = Q\left[\frac{1}{h_i A_i} + \frac{1}{(kA_w/x_w)} + \frac{1}{h_o A_o}\right]$$

$$\therefore \qquad (T_1 - T_4) = Q\left[\frac{1}{h_i A_i} + \frac{1}{(kA_w/x_w)} + \frac{1}{h_o A_o}\right]$$

where T1 and T4 are the average temperatures of the hot and cold fluid respectively.
Therefore equations similar to Equation-1 in terms of overall heat transfer coefficients can be written as

$$Q = U_i A_i (T_1 - T_4)$$
$$Q = U_o A_o (T_1 - T_4)$$

where Ui and Uo are the overall heat transfer coefficients based on the inside area and outside area, respectively.

$$\frac{1}{U_o A_o} = \frac{1}{h_i A_i} + \frac{1}{(kA_w/x_w)} + \frac{1}{h_o A_o}$$

$$\frac{1}{U_o} = \frac{1}{h_i}\left(\frac{A_o}{A_i}\right) + \frac{x_w}{k}\left(\frac{A_o}{A_w}\right) + \frac{1}{h_o}$$

$$\frac{1}{U_o} = \frac{1}{h_i}\left(\frac{D_o}{D_i}\right) + \frac{x_w \cdot D_o}{k\, D_w} + \frac{1}{h_o}$$

$$\frac{1}{U_i} = \frac{1}{h_i} + \frac{x_w D_i}{k \cdot D_w} + \frac{1}{h_o}\left(\frac{D_i}{D_o}\right)$$

For thin walled tubes, the inside and outside radii are not much different from each other, and hence the overall heat transfer coefficient Uo or Ui may be replaced simply by 'U'

$$\boxed{\begin{aligned}\frac{1}{U} &= \frac{1}{h_i} + \frac{x_w}{k} + \frac{1}{h_o}\\ \frac{1}{U} &= \frac{1}{h_i} + \frac{1}{(k/x_w)} + \frac{1}{h_o}\end{aligned}}$$

Reference Books:

- Cengel A. Yunnus. "*Heat Transfer – A Practical Approach*", McGraw Hill.

Fouling Factor:

When the heat transfer equipment is put into service, after sometime, scale, dirt and other solids deposit on both sides of the pipe wall, providing two more resistances to the heat flow. The added resistances must be taken into account in the calculation of the overall heat transfer coefficient. The additional resistances reduce the original value of U and thus the required amount of heat is no longer transferred by the original heat transfer surface. Hence, heat transfer equipments are designed by taking into account the deposition of dirt and scale by introducing a resistance Rd known as the fouling factor (it is a thermal resistance due to scale).

$$\frac{1}{U} = \frac{1}{h_i} + \frac{x_w}{k} + \frac{1}{h_o} + R_d$$

The overall heat transfer coefficient calculated by taking into account Rd is known as the 'design or dirty overall heat transfer coefficient' and the one calculated without taking into account the term Rd is known as the 'clean overall heat transfer coefficient.

Magnitude of film heat transfer coefficients

Table 3.1 : Magnitudes of film heat transfer coefficients (h_i or h_o)

Type of Processes	Range of values of h in W/(m²·K)
No phase change :	
Water (heating or cooling)	300 – 20,000
Gases (heating or cooling)	20 – 300
Air (heating or cooling)	1 – 50
Oils (heating or cooling)	50 – 1500
Organic solvents (heating or cooling)	350 – 3000
Condensing :	
Condensing steam (film-type condensation)	6000 – 20,000
Condensing organic vapours	1000 – 2000
Ammonia	3000 – 6000
Condensing steam (drop-wise condensation)	30,000 – 100,000
Evaporation :	
Water	2000 – 12000
Organic solvents	600 – 2000
Ammonia	1100 – 2300

Table 3.2 gives the range of values of the overall heat transfer coefficients for various fluid systems in a shell and tube equipment.

Reference Books:

- Cengel A. Yunnus. "*Heat Transfer – A Practical Approach*", McGraw Hill.

Table 3.2 : Approximate range of values of overall heat transfer coefficients, U

Hot side (Hot fluid)	Cold side (Cold fluid)	Overall U in W/(m²·K)
Heat exchangers (without phase change) :		
Water	Water	900 – 1700
Gases	Water	20 – 300
Organic solvents	Water	300 – 900
Water	Brine	600 – 1200
Gases	Brine	20 – 300
Heavy organics	Heavy organics	50 – 300
Heaters :		
Steam	Water	1500 – 4000
Steam	Gases	30 – 300
Dowtherm	Gases	20 – 200
Steam	Light oils	300 – 900
Steam	Organic solvents	600 – 1200
Evaporators :		
Steam	Water	2000 – 4000
Steam	Organic solvents	600 – 1200
Steam	Light oils	400 – 1000
Water	Refrigerants	400 – 900
Condensers :		
Steam	Water	2000 – 4000
Saturated organic solvents	Water	600 – 1200
Low boiling hydrocarbons	Water	400 – 1200
Organic solvents with high noncondensable	Water	100 – 500

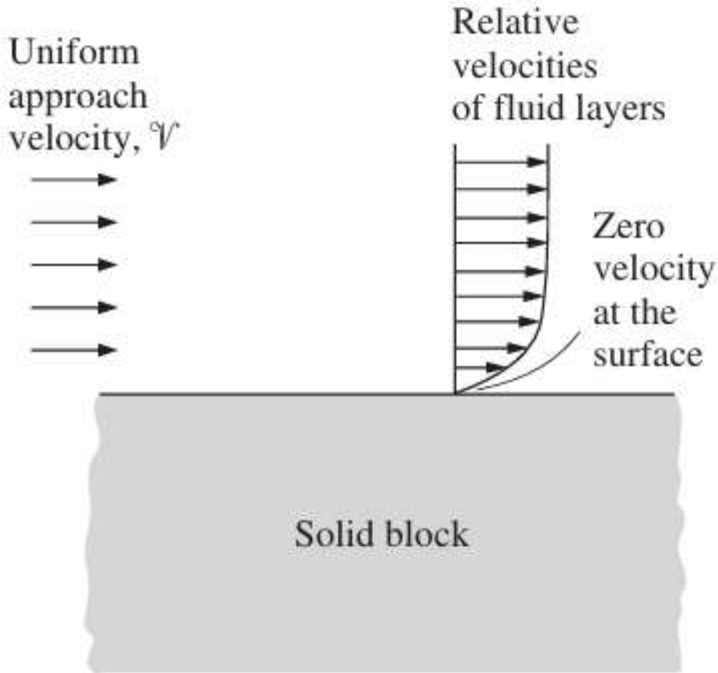
Sensible heat: It is the heat that must be transferred either to raise or lower the temperature of a substance or mixture of substances. It is the heat added or removed from a system to increase or decrease the temperature of the system without changing its phase. The sensible heat transfer is given by $Q = m\ Cp\ \Delta T$ or $\cdot\ m\ Cp\ \Delta T$

Latent heat: When a matter undergoes a phase change, the enthalpy change associated with unit amount of matter at constant temperature and pressure is known as the latent heat of phase change. The latent heat transfer is given by $Q = m\lambda$ where m is the mass and · m is the mass flow rate.

No-slip condition:

- A fluid flowing over a stationary surface comes to a complete stop at the surface because of the no-slip condition.
- When a fluid is forced to flow over a solid surface that is nonporous (i.e., impermeable to the fluid), it is observed that the fluid in motion comes to a complete stop at the surface and assumes a zero velocity relative to the surface.

- That is, the fluid layer in direct contact with a solid surface "sticks" to the surface and there is no slip. In fluid flow, this phenomenon is known as the no-slip condition, and it is due to the viscosity of the fluid
- The no-slip condition is responsible for the development of the velocity profile for flow. Because of the friction between the fluid layers, the layer that sticks to the wall slows the adjacent fluid layer, which slows the next layer, and so on.
- A consequence of the no-slip condition is that all velocity profiles must have zero values at the points of contact between a fluid and a solid.



No-temperature-jump condition:

- A similar phenomenon occurs for the temperature. When two bodies at different temperatures are brought into contact, heat transfer occurs until both bodies assume the same temperature at the point of contact.
- Therefore, a fluid and a solid surface will have the same temperature at the point of contact. This is known as no-temperature-jump condition.
- An implication of the no-slip and the no-temperature jump conditions is that heat transfer from the solid surface to the fluid layer adjacent to the surface is by pure conduction, since the fluid layer is motionless, and can be expressed as

$$\dot{q}_{\text{conv}} = \dot{q}_{\text{cond}} = -k_{\text{fluid}} \left. \frac{\partial T}{\partial y} \right|_{y=0}$$

- where T represents the temperature distribution in the fluid and $(\partial T/\partial y)_{y=0}$ is the temperature gradient at the surface. This heat is then convected away from the surface as a result of fluid motion.

Reference Books:

- Cengel A. Yunnus. "*Heat Transfer – A Practical Approach*", McGraw Hill.

$$h = \frac{-k_{\text{fluid}}(\partial T/\partial y)_{y=0}}{T_s - T_\infty}$$

VELOCITY BOUNDARY LAYER

Consider the parallel flow of a fluid over a flat plate. The x-coordinate is measured along the plate surface from the leading edge of the plate in the direction of the flow, and y is measured from the surface in the normal direction.

The fluid approaches the plate in the x-direction with a uniform upstream velocity of v, which is practically identical to the free-stream velocity u_∞ over the plate away from the surface.

The velocity of the particles in the first fluid layer adjacent to the plate becomes zero because of the no-slip condition. This motionless layer slows down the particles of the neighboring fluid layer as a result of friction between the particles of these two adjoining fluid layers at different velocities.

This fluid layer then slows down the molecules of the next layer, and so on. Thus, the presence of the plate is felt up to some normal distance from the plate beyond which the free-stream velocity u_∞ remains essentially unchanged. As a result, the x-component of the fluid velocity, u will vary from 0 at y = 0 to nearly u_∞ at y = δ.

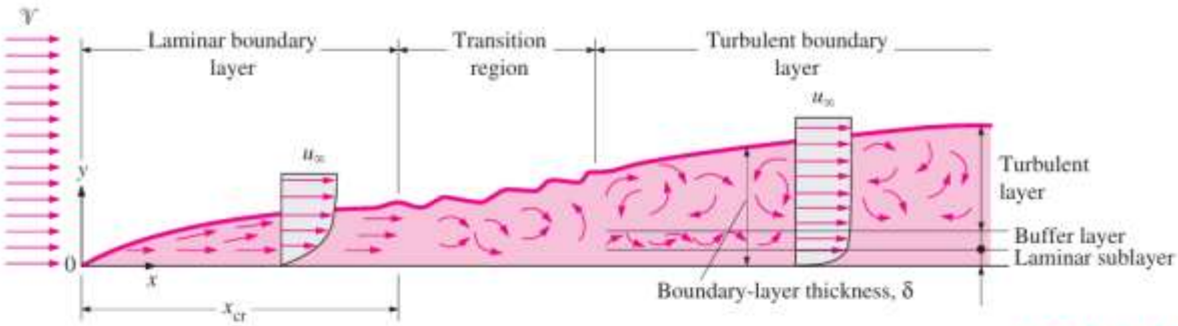


FIGURE 6–10
The development of the boundary layer for flow over a flat plate, and the different flow regimes.

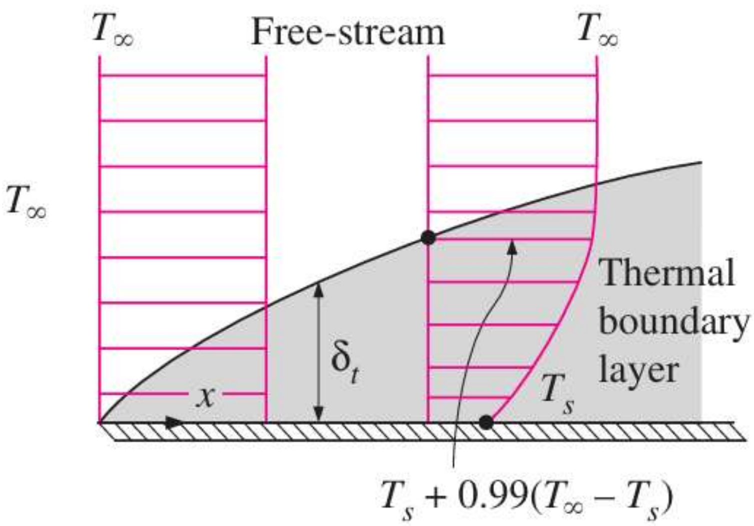
The region of the flow above the plate bounded by in which the effects of the viscous shearing forces caused by fluid viscosity are felt is called the velocity boundary layer.

The boundary layer thickness δ is typically de fined as the distance y from the surface at which u = 0.99u_∞.

The hypothetical line of u = 0.99u_∞ divides the flow over a plate into two regions: the boundary layer region, in which the viscous effects and the velocity changes are significant, and the inviscid flow region, in which the frictional effects are negligible and the velocity remains essentially constant.

THERMAL BOUNDARY LAYER

- Similar to the velocity boundary layer, a thermal boundary layer develops when a fluid at a specified temperature flows over a surface that is at a different temperature.
- Consider the flow of a fluid at a uniform temperature of T over an isothermal flat plate at temperature Ts.
- The fluid particles in the layer adjacent to the surface will reach thermal equilibrium with the plate and assume the surface temperature Ts.
- These fluid particles will then exchange energy with the particles in the adjoining-fluid layer, and so on.
- As a result, a temperature profile will develop in the flow field that ranges from Ts at the surface to T∞ sufficiently far from the surface.
- The flow region over the surface in which the temperature variation in the direction normal to the surface is significant is the thermal boundary layer.
- The thickness of the thermal boundary layer δ_t at any location along the surface is defined as the distance from the surface at which the temperature difference T-Ts equals 0.99(T∞-Ts).
- **Note that for the special case of Ts=0, we have T = 0.99T∞ at the outer edge of the thermal boundary layer, which is analogous to u = 0.99u∞ for the velocity boundary layer.**
- The thickness of the thermal boundary layer increases in the flow direction, since the effects of heat transfer are felt at greater distances from the surface further down stream.
- The convection heat transfer rate anywhere along the surface is directly related to the temperature gradient at that location.
- **Therefore, the shape of the temperature profile in the thermal boundary layer dictates the convection heat transfer between a solid surface and the fluid flowing over it.**
- In flow over a heated (or cooled) surface, both velocity and thermal boundary layers will develop simultaneously.
- **Noting that the fluid velocity will have a strong influence on the temperature profile, the development of the velocity boundary layer relative to the thermal boundary layer will have a strong effect on the convection heat transfer.**



Reference Books:

- Cengel A. Yunnus. "*Heat Transfer – A Practical Approach*", McGraw Hill.

Dimensionless numbers

1. **Reynolds Number**

- After exhaustive experiments in the 1880s, Osborn Reynolds discovered that the flow regime depends mainly on the ratio of the inertia forces to viscous forces in the fluid.
- This ratio is called the Reynolds number, which is a dimensionless quantity, and is expressed for external flow, where V is the upstream velocity (equivalent to the free-stream velocity u for a flat plate), Lc is the characteristic length of the geometry, and / is the kinematic viscosity of the fluid.

$$\mathrm{Re} = \frac{\text{Inertia forces}}{\text{Viscous}} = \frac{\mathcal{V} L_c}{\nu} = \frac{\rho \mathcal{V} L_c}{\mu}$$

- The Reynolds number at which the flow becomes turbulent is called the critical Reynolds number. The value of the critical Reynolds number is different for different geometries.
- For flow over a flat plate, the generally accepted value of the critical Reynolds number is $\mathrm{Re}_{cr} = 5 \times 10^5$, where xcr is the distance from the leading edge of the plate at which transition from laminar to turbulent flow occurs.

$$\mathrm{Re}_{\mathrm{cr}} = \mathcal{V} x_{\mathrm{cr}}/\nu = u_\infty x_{\mathrm{cr}}/\nu$$

2. **Prandtl Number**

- The relative thickness of the velocity and the thermal boundary layers is best described by the dimensionless parameter Prandtl number, defined as

$$\mathrm{Pr} = \frac{\text{Molecular diffusivity of momentum}}{\text{Molecular diffusivity of heat}} = \frac{\nu}{\alpha} = \frac{\mu C_p}{k}$$

- It is named after Ludwig Prandtl, who introduced the concept of boundary layer in 1904 and made significant contributions to boundary layer theory.
- The Prandtl numbers of fluids range from less than 0.01 for liquid metals to more than 100,000 for heavy oils.
- Prandtl number is in the order of 10 for water.
- The Prandtl numbers of gases are about 1, which indicates that both momentum and heat dissipate through the fluid at about the same rate.
- Heat diffuses very quickly in liquid metals ($\mathrm{Pr} \ll 1$) and very slowly in oils ($\mathrm{Pr} \gg 1$) relative to momentum.
- Consequently the thermal boundary layer is much thicker for liquid metals and much thinner for oils relative to the velocity boundary layer.

TABLE 6–2

Typical ranges of Prandtl numbers for common fluids

Fluid	Pr
Liquid metals	0.004–0.030
Gases	0.7–1.0
Water	1.7–13.7
Light organic fluids	5–50
Oils	50–100,000
Glycerin	2000–100,000

3. Nusselt Number

- To nondimensionalize the heat transfer coefficient h with the Nusselt number

$$\mathrm{Nu} = \frac{hL_c}{k}$$

- where k is the thermal conductivity of the fluid and Lc is the characteristic length.
- The Nusselt number is named after Wilhelm Nusselt, who made significant contributions to convective heat transfer in the first half of the twentieth century.
- The Nusselt number represents the enhancement of heat transfer through a fluid layer as a result of convection relative to conduction across the same fluid layer.
- **The larger the Nusselt number, the more effective the convection.**
- A Nusselt number of Nu = 1 for a fluid layer represents heat transfer across the layer by pure conduction.

$$\dot{q}_{\text{conv}} = h\Delta T$$

$$\dot{q}_{\text{cond}} = k\frac{\Delta T}{L}$$

Taking their ratio gives

$$\frac{\dot{q}_{\text{conv}}}{\dot{q}_{\text{cond}}} = \frac{h\Delta T}{k\Delta T/L} = \frac{hL}{k} = \mathrm{Nu}$$

Reference Books:

- Cengel A. Yunnus. "*Heat Transfer – A Practical Approach*", McGraw Hill.

$$\text{Reynolds number} = \frac{\text{Inertia forces}}{\text{Viscous forces}}$$

$$\text{Grashof number} = \frac{\text{Buoyancy forces} \times \text{inertia forces}}{(\text{viscous forces})^2}$$

$$\text{Prandtl number} = \frac{\text{molecular diffusivity of momentum}}{\text{molecular diffusivity of heat}}$$

$$\text{Nusselt number} = \frac{\text{Wall heat transfer rate}}{\text{Heat transfer by conduction}}$$

Application of Dimensional Analysis to heat transfer by convection

In convection studies, it is common practice to nondimensionalize the governing equations and combine the variables, which group together into dimensionless numbers in order to reduce the number of total variables.

Application of dimensional analysis for a forced convection equation

To derive a relationship for heat transfer coefficient, h for forced convection heat transfer on the assumption that the coefficient h is a function of the following variables: It is given that h is a function of L, u, ρ, μ, Cp and k

$$h = f(\rho, u, L, \mu, C_p, k)$$

$$h = A(\rho^a u^b L^c \mu^d C_p^e k^f)$$

: A, a, b, c, d, e, and f are constants.

Express the energy terms mechanically by dimensions of the variables :

The dimensions of each variable in terms of M, L, T, θ and F are :

Parameter	Units	Fundamental dimensions
h	$N.m/(m^2 \cdot s \cdot K) = W/(m^2 \cdot K)$	$FL/L^2\theta T = FL^{-1}\theta^{-1}T^{-1}$
k	$N \cdot m/(m \cdot s \cdot K) = W/(m \cdot K)$	$FL/L\theta T = F\theta^{-1}T^{-1}$
C_p	$N \cdot m/(kg \cdot K) = J/(kg \cdot K)$	$FL/MT = FLM^{-1}T^{-1}$
μ	$kg/(m \cdot s)$	$M/L\theta = ML^{-1}\theta^{-1}$
u	m/s	$L/\theta = L\theta^{-1}$
l	m	$L = L$
ρ	kg/m^3	$M/L^3 = ML^{-3}$

Substituting the dimensions of all parameters in the above equation

$$FL^{-1}\theta^{-1}T^{-1} = A(ML^{-3})^a (L\theta^{-1})^b (L)^c (ML^{-1}\theta^{-1})^d (FLM^{-1}T^{-1})^e (F\theta^{-1}T^{-1})^f$$

Equating the indices/powers of M, L, θ, T and F, on LHS with those on RHS of the above equation gives

$$
\begin{aligned}
\text{Force, F} &: \quad 1 = e + f \\
\text{Mass, M} &: \quad 0 = a + d - e \\
\text{Length, L} &: \quad -1 = -3a + b + c - d + e \\
\text{Temperature, T} &: \quad -1 = -e - f \\
\text{Time, } \theta &: \quad -1 = -b - d - f
\end{aligned}
$$

obtain the constants c, d, etc. in terms of a and e

From Equation (i),

$$f = 1 - e$$

From Equation (ii),

$$d = e - a$$

From Equation (v),

$$
\begin{aligned}
-1 &= -b - d - f \\
b &= 1 - d - f = 1 - (e - a) - f \\
b &= 1 - e + a - f \\
b &= a + (1 - e) - f = a + f - f \\
\therefore \quad b &= a
\end{aligned}
$$

From Equation (iii),

$$
\begin{aligned}
-1 &= -3a + b + c - d + e \\
c &= 3a - b + d - e - 1 \\
&= 3a - a + (e - a) - e - 1 \\
c &= 2a - a - 1 = a - 1
\end{aligned}
$$

$$h = A (\rho)^a (u)^a (L)^{a-1} (\mu)^{e-a} (C_p)^e \ (k)^{1-e}$$

Collecting the terms, we get

$$h = A \left(\frac{Lu\rho}{\mu}\right)^a \left(\frac{C_p \mu}{k}\right)^e (k/L)$$

$$\therefore \qquad \frac{hL}{k} = A \left(\frac{Lu\rho}{\mu}\right)^a \left(\frac{C_p \mu}{k}\right)^e$$

In the case of pipe, the linear dimension is D (inside diameter) and the above equation becomes

$$\boxed{\frac{hD}{k} = A \left(\frac{Du\rho}{\mu}\right)^a \left(\frac{C_p \mu}{k}\right)^e}$$

This equation contains three dimensionless groups/numbers :

(i) Nusselt number, NNu = hL/k
(ii) Reynolds number, NRe = Luρ/μ
(iii) Prandtl number, NPr = Cp μ/k.

Application of dimensional analysis for a natural convection equation

coefficient h is a function of l, k, ρ, Cp, μ, βg and ΔT.

$$h = f\,[l, \rho, \mu, k, C_p, \beta\cdot g, \Delta T]$$

$$h = \alpha\,[l^a \rho^b \mu^c k^d C_p{}^e (\beta g)^f (\Delta T)^g]$$

β·g – the product of the acceleration due to gravity and coefficient of cubical expansion of the fluid

Reference Books:

- Cengel A. Yunnus. "*Heat Transfer – A Practical Approach*", McGraw Hill.

The dimensions of each variable in terms of M, L, T, F and θ are :

Parameter	Units	Fundamental dimensions
h	$N.m/(m^2 \cdot s \cdot K) = W/(m^2 \cdot K)$	$FL/L^2\theta T = FL^{-1}\theta^{-1}T^{-1}$
k	$N \cdot m/(m \cdot s \cdot K) = W/(m \cdot K)$	$FL/L\theta T = F\theta^{-1}T^{-1}$
C_p	$N \cdot m/(kg \cdot K) = J/(kg \cdot K)$	$FL/MT = FLM^{-1}T^{-1}$
μ	$kg/(m \cdot s)$	$M/L\theta = ML^{-1}\theta^{-1}$
l	m	$L = L$
ρ	kg/m^3	$M/L^3 = ML^{-3}$
βg	$m/(s^2 \cdot K)$	$L/\theta^2 T = L\theta^{-2}T^{-1}$
ΔT	K	$T = T$

Substituting the dimensions of each variable in the above equation, we get :

$$FL^{-1}\theta^{-1}T^{-1} = \alpha (L)^a (ML^{-3})^b (ML^{-1}\theta^{-1})^c (F\theta^{-1}T^{-1})^d (FLM^{-1}T^{-1})^e (L\theta^{-2}T^{-1})^f \times (T)^g$$

Equating the indices/powers on the LHS and RHS of the above equation in force, mass

Force, $\quad F : 1 = d + e$

Mass, $\quad M : 0 = b + c - e$

Length, $\quad L : -1 = a - 3b - c + e + f$

Temperature, $\quad T : -1 = -d - e - f + g$

Time, $\quad \theta : -1 = -c - d - 2f$

Let us obtain the constants a, b, c and d in terms of e and f. From Equation (i),

$$d = 1 - e$$

From Equation (v),

$$c = 1 - d - 2f, \quad \text{but } d = 1 - e$$

$$\therefore \quad c = 1 - (1 - e) - 2f = e - 2f$$

From Equation (ii), we have

$$b = -c + e \quad \text{but } c = e - 2f$$

$$\therefore \quad b = -(e - 2f) + e = 2f$$

From Equation (iii), we have

$$a = -1 + 3b + c - e - f$$

$$a = -1 + 3(2f) + (e - 2f) - e - f = -1 + 3f$$

From Equation (iv), we have

$$g = -1 + d + e + f = -1 + 1 + f = f \quad [\text{as } d + e = 1]$$

Substituting back the values of constants a, b, c and d, the equation for h becomes

$$h = \alpha\, (l)^{-1+3f}\, (\rho)^{2f}\, (\mu)^{e-2f}\, (k)^{1-e}\, (C_p)^{e}\, (\beta g)^{f}\, (\Delta T)^{f}$$

Collecting the terms, we get

$$h = \alpha \left(\frac{C_p \mu}{k}\right)^{e} \left(\frac{l^3 \rho^2 \beta g\, \Delta T}{\mu^2}\right)^{f} (k/L)$$

$$\boxed{\frac{hL}{k} = \alpha \left(\frac{C_p\, \mu}{k}\right)^{e} \left(\frac{l^3 \rho^2 \beta g\, \Delta T}{\mu^2}\right)^{f}}$$

$\dfrac{C_p\, \mu}{k}$ is the Prandtl number (N_{Pr})

hL/k is the Nusselt number (N_{Nu})

$\dfrac{l^3 \rho^2 \beta \cdot g\, \Delta T}{\mu^2}$ is the Grashof number (N_{ar})

SOLUTIONS OF CONVECTION EQUATIONS FOR A FLAT PLATE

Consider laminar flow of a fluid over a flat plate.

The x-coordinate is measured along the plate surface from the leading edge of the plate in the direction of the flow, and y is measured from the surface in the normal direction.

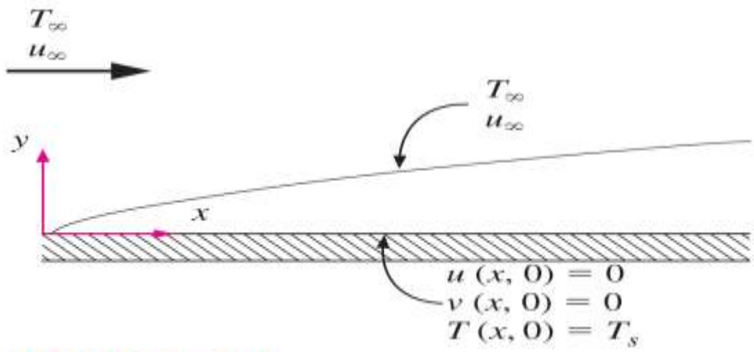


FIGURE 6–26
Boundary conditions for flow over a flat plate.

The fluid approaches the plate in the x-direction with a uniform upstream velocity, which is equivalent to the free stream velocity u∞.

When viscous dissipation is negligible, the continuity, momentum, and en ergy equations reduce for steady, incompressible, laminar flow of a fluid with constant properties over a flat plate to

Continuity: $$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

Momentum: $$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \nu\frac{\partial^2 u}{\partial y^2}$$

Energy: $$u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} = \alpha\frac{\partial^2 T}{\partial y^2}$$

with the boundary conditions

At $x = 0$: $u(0, y) = u_\infty$, $T(0, y) = T_\infty$

At $y = 0$: $u(x, 0) = 0$, $v(x, 0) = 0$, $T(x, 0) = T_s$

As $y \to \infty$: $u(x, \infty) = u_\infty$, $T(x, \infty) = T_\infty$

When fluid properties are assumed to be constant and thus independent of temperature, the first two equations can be solved separately for the velocity components u and v.

Blasius was also aware from the work of Stokes that is proportional to $\sqrt{\nu x/u_\infty}$, and thus he defined a dimensionless similarity variable as

$$\eta = y\sqrt{\frac{u_\infty}{\nu x}}$$

The nondimensional velocity profile can be obtained by plotting u/u∞ against η.

The value of η corresponding to u/u 0.992 is 5.0. Substituting 5.0 and y into the definition of the similarity variable gives

$$5.0 = \delta\sqrt{u_\infty/\nu x}$$

Then the velocity boundary layer thickness becomes

$$\boxed{\delta = \frac{5.0}{\sqrt{u_\infty/\nu x}} = \frac{5.0x}{\sqrt{\mathrm{Re}_x}}}$$

The shear stress on the wall

$$\tau_w = \mu \left.\frac{\partial u}{\partial y}\right|_{y=0}$$

$$\boxed{\tau_w = 0.332 u_\infty \sqrt{\frac{\rho\mu u_\infty}{x}} = \frac{0.332\rho u_\infty^2}{\sqrt{\mathrm{Re}_x}}}$$

The local convection coefficient and Nusselt number

$$\boxed{h_x = \frac{q_s}{T_s - T_\infty} = \frac{-k(\partial T/\partial y)|_{y=0}}{T_s - T_\infty} = 0.332\ \mathrm{Pr}^{1/3} k\sqrt{\frac{u_\infty}{\nu x}}}$$

$$\boxed{\mathrm{Nu}_x = \frac{h_x x}{k} = 0.332\ \mathrm{Pr}^{1/3}\mathrm{Re}_x^{1/2} \qquad \mathrm{Pr} > 0.6}$$

Solving Eq. numerically for the temperature profile for different Prandtl numbers, and using the definition of the thermal boundary layer, it is determined that $\delta/\delta t \cong \mathrm{Pr}^{1/3}$.

Then the thermal boundary layer thickness becomes

$$\boxed{\delta_t = \frac{\delta}{\mathrm{Pr}^{1/3}} = \frac{5.0x}{\mathrm{Pr}^{1/3}\sqrt{\mathrm{Re}_x}}}$$

Empirical equations for calculation of heat transfer coefficients

Film coefficients-in pipes – Laminar flow

Heat transfer to laminar flow may be occurred whenever fluids to be heated or cooled are viscous or flow rates are low. In polymer, food or pharmaceutical industry, we may come across a heat exchanger where the flow is in the laminar region (NRe < 2100).

For heating or cooling of viscous fluids, an empirical correlation is the **Sieder-Tate equation** for the calculation of the heat transfer coefficient for laminar flow of fluids in horizontal tubes or pipes.

Equation is valid for

NRe < 2100 and (NRe NPr , D/L) > 100

The term (μ/μw)0.14 is called as the Sider-Tate correction. The term (μ/μw) is the ratio of the viscosity of the fluid at the bulk temperature and the viscosity of the fluid at the wall temperature.

$$N_{Nu} = 1.86\,[(N_{Re})\,(N_{Pr})\,(D/L)]^{1/3}\,[\mu/\mu_w]^{0.14}$$

$$\frac{hD}{k} = 1.86\left[\left(\frac{Du\rho}{\mu}\right)\left(\frac{C_p\,\mu}{k}\right)\left(\frac{D}{L}\right)\right]^{1/3}\left(\frac{\mu}{\mu_w}\right)^{0.14}$$

Film coefficients - in pipes – Turbulent flow

he most important situation in heat transfer is the flow of heat in a stream of fluid in turbulent flow in pipes or tubes. Turbulence is encountered at NRe > 2100. As the rate of heat transfer is greater in turbulent flow than in laminar flow, most of the heat transfer equipments are operated in the turbulent range. The Dittus-Boelter equation for predicting the heat transfer coefficient for turbulent flow in tubes or pipes is

$$\frac{hD}{k} = 0.023\left[\frac{Du\rho}{\mu}\right]^{0.8}\left[\frac{C_p\,\mu}{k}\right]^{a}$$

where 'a' has a value of 0.4 for heating and 0.3 for cooling. It is commonly used for water like materials.

For heating :

$$N_{Nu} = 0.023\ [N_{Re}]^{0.8}\ [N_{Pr}]^{0.4}$$

For cooling :

$$N_{Nu} = 0.023\ [N_{Re}]^{0.8}\ [N_{Pr}]^{0.3}$$

In the Colburn equation, the Stanton number (Nst) is used instead of the Nusselt number (NNu):

$$N_{st} = \frac{h}{C_p\,\rho u}$$

$$\frac{h}{C_p\,\rho u} = \frac{hD}{k} \cdot \frac{\mu}{Du\rho} \cdot \frac{k}{C_p\,\mu}$$

$$N_{st} = N_{Nu} \cdot N_{Re}^{-1} \cdot N_{Pr}^{-1}$$

$$N_{Nu} = N_{st} \cdot N_{Re} \cdot N_{Pr}$$

The Colburn analogy; Colburn j factor

For NRe between 5000 – 200000, the friction factor (f) for a smooth pipe is given by the following empirical equation

$$f = 0.046 \left(\frac{Du\rho}{\mu}\right)^{-0.2}$$

Comparison of Equation for heat transfer in turbulent flow inside a pipe/tube gives

$$\frac{h}{C_p\,\rho u}\left(\frac{C_p\,\mu}{k}\right)^{2/3}\left(\frac{\mu_w}{\mu}\right)^{0.14} \equiv j_H = \frac{f}{2}$$

Equation is the Colburn analogy between heat transfer and fluid friction. The factor jH is called the Colburn j factor. Equation can be written in the j-factor form as given below : jH = 0.

Wilson Plot:

This plot is used to determine the film heat transfer coefficients. Consider a shell and tube heat exchanger wherein steam is condensing on the shell side and a cold fluid is flowing through the tubes in the turbulent flow region. T

he overall heat transfer coefficient is determined by the direct measurements of heat transfer rate, overall temperature difference and area at various cold fluid velocities. In this case, the condensing steam side coefficient (ho) remains almost constant and the resistance offered by the metal wall is also constant. Assuming clean tubes, Equation (3.19) reduces to

$$\frac{1}{U} = \frac{1}{h_i} + C$$

for turbulent flow

$$N_{Nu} \propto (N_{Re})^{0.8}$$

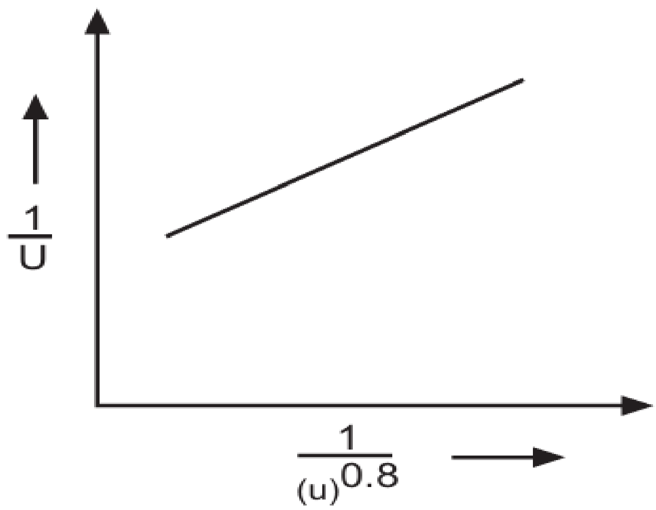
$$h_i \propto (u)^{0.8}$$

$$h_i = a\,(u)^{0.8}$$

Thus,

$$\frac{1}{U} = \frac{1}{a\,(u)^{0.8}} + C$$

where 'u' is the linear velocity of the cold fluid. A plot of 1/U v/s 1/(u)^0.8 results in a straight line with a slope equal to 1/a and an intercept equal to xw/k + 1/ho . The value of ho is obtained from the intercept and 'a' represents the value of film coefficient hi for a unit velocity of the cold fluid.' Such a plot shown in Fig is known as the Wilson plot.



Example 3.6 : *Calculate the heat transfer coefficient for fluid flowing through a tube having inside diameter 40 mm at a rate of 5500 kg/h. Assume that the fluid is being heated.*

Data : *Properties of fluid at mean bulk temperature :*

Viscosity of flowing fluid = 0.004 (N.s)/m²

Density of flowing fluid = 1.07 g/cm³

Specific heat of flowing fluid = 2.72 kJ/(kg.K)

Thermal conductivity of flowing fluid = 0.256 W/(m·K)

Make use of Dittus – Boelter equation.

Solution : Basis : 5500 kg/h of fluid flowing through the tube

$$\text{Mass flow rate} = \dot{m} = 5500 \text{ kg/h} = \frac{5500}{3600} = 1.53 \text{ kg/s}$$

$$\text{Density of the fluid} = \rho = 1.07 \text{ g/cm}^3 = 1070 \text{ kg/m}^3$$

$$\text{Volumetric flow rate} = \dot{m}/\rho = \frac{1.53}{1070} = 1.43 \times 10^{-3} \text{ m}^3\text{/s}$$

$$\text{Cross-sectional area of the tube} = \frac{\pi}{4} D_i^2$$

where $D_i = 40 \text{ mm} = 0.04 \text{ m}$

$$A = \frac{\pi}{4} (0.040)^2 = 1.256 \times 10^{-3} \text{ m}^2$$

Let us calculate u.

$$\text{Velocity of the flowing fluid} = u = \frac{1.43 \times 10^{-3}}{1.256 \times 10^{-3}} = 1.14 \text{ m/s}$$

Let us calculate N_{Re}.

$$\text{Reynolds number} = N_{Re} = \frac{D_i u \rho}{\mu}$$

where D_i = inside diameter of pipe

$= 40 \text{ mm} = 0.04 \text{ m}$

$\rho = 1070 \text{ kg/m}^3$

$u = 1.14 \text{ m/s}$

μ = Viscosity

$= 0.004 \text{ (N.s)/m}^2 = 0.004 \text{ kg/(m·s)}$

$$N_{Re} = \frac{0.04 \times 1.14 \times 1070}{0.004} = 12198$$

Reference Books:

- Cengel A. Yunnus. "*Heat Transfer – A Practical Approach*", McGraw Hill.

Let us calculate N_{Pr}.

$$N_{Pr} = \frac{C_p \cdot \mu}{k}$$

$$C_p = 2.72 \text{ kJ/(kg·K)} = 2.72 \times 10^3 \text{ J/(kg·K)}$$

$$k = 0.256 \text{ W/(m·K)}$$

$$\mu = 0.004 \text{ kg/(m·s)}$$

$$N_{Pr} = \frac{2.72 \times 10^3 \times 0.004}{0.256} = 42.5$$

As N_{Re} is greater than 10000, the flow is turbulent.

Let us obtain h_i.

The empirical relationship for turbulent flow, i.e., the Dittus–Boelter equation for heating is

$$N_{Nu} = 0.023\ (N_{Re})^{0.8}\ (N_{Pr})^{0.4}$$

$$N_{Nu} = 0.023\ (12198)^{0.8}\ (42.5)^{0.4} = 191.5$$

N_{Nu} = Nusselt number

$$\frac{h_i D_i}{k} = 191.5$$

where, h_i = inside heat transfer coefficient

D_i = inside diameter of pipe = 0.04 m

k = thermal conductivity of fluid = 0.256 W/(m·K)

$$\frac{h_i\ (0.04)}{0.256} = 191.5$$

$$h_i = \mathbf{1225.5\ W/(m^2 \cdot K)}$$

Example 3.12 : *Find the inside heat transfer coefficient using Sieder-Tate equation for turbulent flow.*

Data : *I.d. of tube = 20 mm,* $N_{Re} = 15745$

Viscosity of fluid at bulk mean temperature $= 550 \times 10^{-5}$ *Pa·s*

Viscosity of fluid at average wall temperature $= 900 \times 10^{-6}$ *Pa·s*

Prandtl number = 36, Thermal conductivity of fluid = k = 0.25 W/(m·K)

Solution : The Sieder-Tate equation is

$$\frac{h_i D_i}{k} = 0.023 (N_{Re})^{0.8} (N_{Pr})^{1/3} \left(\frac{\mu}{\mu_w}\right)^{0.14}$$

where, $\left(\frac{\mu}{\mu_w}\right)^{0.14}$ = Sieder-Tate correction factor

N_{Re} = Reynolds number = 15745

N_{Pr} = Prandtl number = 36

h_i = inside heat transfer coefficient, W/(m²·K)

D_i = inside diameter of pipe = 20 mm = 0.002 m

k = Thermal conductivity of fluid = 0.25 W/(m·K)

μ = 550×10^{-6} Pa·s

μ_w = 900×10^{-6} Pa·s

Substituting the values in Equation (1), we get

$$\frac{h_i (0.02)}{0.25} = 0.023 \times (15745)^{0.8} (36)^{1/3} \times \left(\frac{550 \times 10^{-6}}{900 \times 10^{-6}}\right)^{0.14}$$

$$\frac{h_i (0.02)}{0.25} = 0.023 \times 2278.84 \times 3.3 \times 0.933$$

$$\frac{h_i (0.02)}{0.25} = 161.37$$

$$h_i = 2017$$

Inside heat transfer coefficient = **2017 W/(m²·K)**

Example 3.18 : *Determine the heat transfer coefficient for water flowing in a tube of 16 mm diameter at a velocity of 3 m/s. The temperature of the tube is 297 K (24°C) and the water enters at 353 K (80°C) and leaves at 309 K (36°C). Using (i) Dittus - Boelter equation and (ii) Sieder-Tate equation.*

Data : *Properties of water at 331 K (58°C), i.e., at arithmetic mean-bulk temperature are :* $\rho = 984.1$ kg/m³, $C_p = 4187$ J/(kg·K), $\mu = 485 \times 10^{-6}$ Pa·s, $k = 0.657$ W/(m·K)

Viscosity of water at 297 K (24°C), $\mu_w = 920 \times 10^{-6}$ Pa·s

Reference Books:

- Cengel A. Yunnus. "*Heat Transfer – A Practical Approach*", McGraw Hill.

Solution : Let us find N_{Re}.

$$N_{Re} = \frac{Du\rho}{\mu}$$

where $D = 16\text{ mm} = 0.016\text{ m}$, $u = 3\text{ m/s}$

$\mu = 485 \times 10^{-6}\text{ Pa}\cdot\text{s}$ or $(\text{N}\cdot\text{s})/\text{m}^2 = 485 \times 10^{-6}\text{ kg/(m}\cdot\text{s)}$

$\rho = 984.1\text{ kg/m}^3$ at arithmetic mean bulk temperature

$$N_{Re} = \frac{0.016 \times 3 \times 984.1}{485 \times 10^{-6}} = 97395$$

Let us find N_{Pr}.

$$N_{Pr} = \frac{C_p\,\mu}{k}$$

where $k = 0.657\text{ W/(m}\cdot\text{K)}$

$C_p = 4187\text{ J/(kg}\cdot\text{K)}$

$$N_{Pr} = \frac{4187 \times 485 \times 10^{-6}}{0.657} = 3.09$$

(i) The Dittus-Boelter equation for cooling is

$$N_{Nu} = 0.023\,(N_{Re})^{0.8}\,(N_{Pr})^{0.3}$$

$$\frac{hD}{k} = 0.023\,(N_{Re})^{0.8}\,(N_{Pr})^{0.3}$$

where h is the film heat transfer coefficient.

$$h = 0.023 \times (N_{Re})^{0.8}\,(N_{Pr})^{0.3} \times \frac{k}{D}$$

$$h = 0.023\,(97395)^{0.8}\,(3.09)^{0.3} \times \left(\frac{0.657}{0.016}\right)$$

$$h = \mathbf{12972.6\ W/(m^2\cdot K)}$$

(ii) The Sieder-Tate equation is

$$N_{Nu} = \frac{hD}{k} = 0.023\,(N_{Re})^{0.8}\,(N_{Pr})^{1/3}\,(\mu/\mu_w)^{0.14}$$

$$h = 0.023\,(N_{Re})^{0.8}\,(N_{Pr})^{1/3}\,(\mu/\mu_w)^{0.14}\,(k/D)$$

where $\mu = 485 \times 10^{-6}\text{ Pa}\cdot\text{s}$, $\mu_w = 920 \times 10^{-6}\text{ Pa}\cdot\text{s}$

$$h = 0.023\,(97395)^{0.8}\,(3.09)^{1/3}\left(\frac{485 \times 10^{-6}}{920 \times 10^{-6}}\right)^{0.14} \times \left(\frac{0.657}{0.016}\right)$$

$$= \mathbf{12267.7\ W/(m^2\cdot K)}$$